Ghazal Abdollahi

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Objective

My research is about how computing systems can notice what is forming in their own data early enough to act on it. I look for early, readable signs in the data systems already produce, so they can act sooner: Interpretable Signal-to-Control (O→S→C). I will graduate with my PhD by March 2027 at the latest and am looking for a tenure-track or postdoctoral position to keep exploring this in new areas.

Education

Fall 2022 – Present **Ph.D. candidate in Computer Science**, *University of Utah*, Salt Lake City, UT
Kahlert School of Computing, Flux member. Advisor: Prof. Robert Ricci.

M.Sc. in Information Technology Engineering – Computer Networks, *Salman Institute of Higher Education*, Mashhad, Iran

B.Sc. in Information Technology Engineering, *Payame Noor University*, Iran

Accepted/Published Papers

- Vertex: Towards Recommender Firewalls Resilient to Post-Detection Evasion (poster, ACM IMC 2026)
- EPIC: When Learners Disagree, Campaigns Speak; Perceiving Structure in Unfolding SSH Episodes (poster, ACM SIGCOMM 2026)
- Combo: Proactive Momentum and the Reactive Wall (ACM NetNeg, SIGCOMM 2026)
- WarmFlex: Coordinated Stabilization of Cold-Start Frequency and Idle Containers in Serverless Platforms (IEEE SmartCloud 2026)
- NOD: Uncovering Intense Attackers' Behavior Through Nested Outlier Detection from SSH Logs (NDSS, PRISM 2026)
- Atomic Resolution Mapping and Electrochemical Analysis of Illicit Drug-Aptamer Complexes and Their Terahertz Signatures (IEEE NMDC 2025)
- FMap: A Fuzzy Map for Scheduling Elephant Flows Through Jumping (Concurrency and Computation)
- Flow-Aware Forwarding in SDN Data Centers Using a Knapsack-PSO-Based Solution (IEEE Transactions on Network and Service Management)

Under Review

- Graspic: Grasping the Momentum of Forming SSH Attack Phenotypes Toward Agentic Defense (AIDC '26 Workshop, ACSAC 2026)
- Ignite: A Comprehensive Runtime Framework for Cold-Start and Idle Container Co-Optimization in Serverless Platforms (IEEE Transactions on Network and Service Management, under revision)
- What Crystallizes When Domain-Specific Semantics Evaporate, and Can Defender Agents See It? (AIDC '26 Workshop, ACSAC 2026)
- Microfabricated Aptamer-Based Wearable Biosensor with Gradient Boosting for Real-Time THC Detection and TinyML Implementation (IEEE MEMS 2027)

Grants & Awards

- N2Women Travel Grant, ACM MobiCom 2026, Austin, TX.
- Conference Travel Grants: ACM RecSys 2026, USENIX PEPR '26, NSDI 2025, KubeCon 2024, and ACM CCS 2024.

Experience

Summer 2024, Summer 2026 — **ML-based Biomedical Detection Intern**, *ECE Department, University of Utah*, Salt Lake City, UT
Supervisor: Prof. Massood Tabib-Azar (AMANDA Lab).

Teaching Assistance

Fall 2026 — **Teaching Assistant**, *CS 4000 – Senior Capstone Design, University of Utah*, Salt Lake City, UT

- Advise senior capstone teams and support grading, project development, and student questions.

Services

- Volunteer Reviewer: IEEE Transactions on Network and Service Management
- Volunteer Reviewer: Future Generation Computer Systems (Elsevier)
- Artifact Evaluation: ACM SIGMETRICS 2026 (shadow reviewer)
- Artifact Evaluation: IEEE ICPE

Key Skills

Technical

- Cloud Computing
- Serverless Computing
- Traffic Scheduling
- Resource Allocation
- Attacker Behavior
- AI/ML for Networking & Security
- Workload Characterization
- Behavioral Signal Study
- Software-Defined Networking (SDN)
- Anomaly & Intrusion Detection
- Interpretable/Explainable AI
- Container Orchestration

References

- Prof. Robert Ricci: ricci@cs.utah.edu
- Prof. Massood Tabib-Azar: azar.m@utah.edu
- Prof. H. James de St. Germain: germain@cs.utah.edu
- Prof. David Bean: david.bean@utah.edu